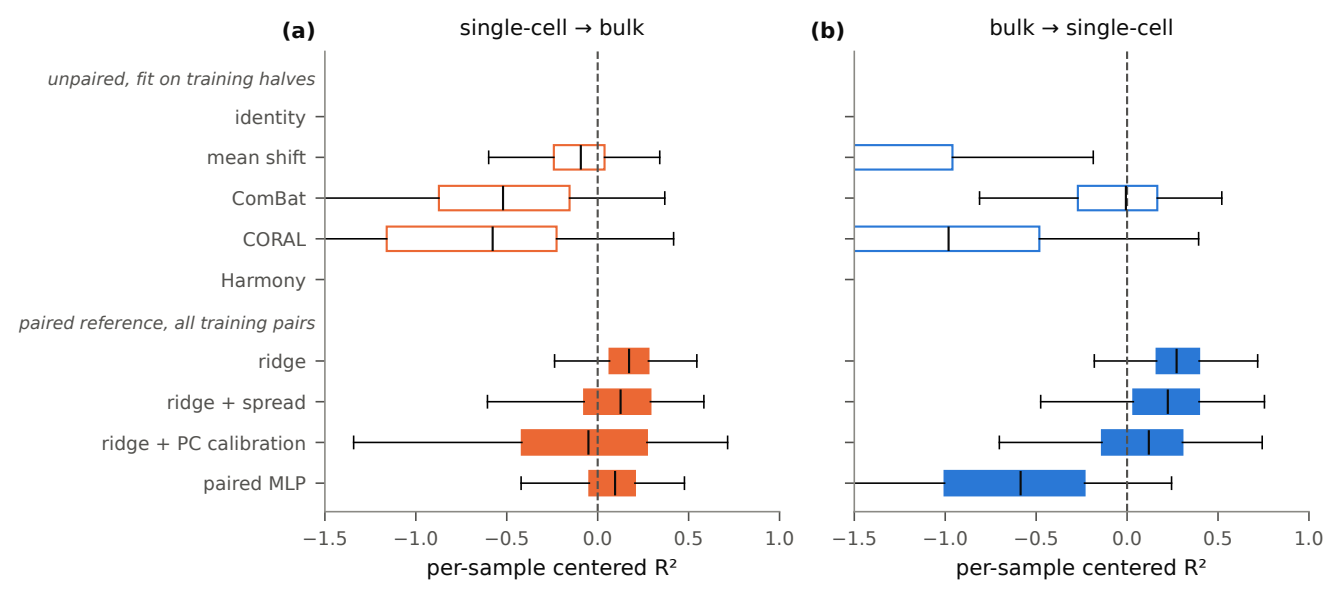
(a)
single-cell → bulk
(b)
bulk → single-cell
unpaired, fit on training halves
identity
mean shift
ComBat
CORAL
Harmony
paired reference, all training pairs
ridge
ridge + spread
ridge + PC calibration
paired MLP
−1.5
−1.0
−0.5
0.0
0.5
1.0
per-sample centered R²